

1. Personal Details & Profiles

- **Full Name:** Somesh Chevula (also recorded as Chevula Somesh)
- **Date of Birth:** April 25, 2004 (Age: 22)
- **Location:** Hyderabad, Telangana, India (Open to relocation/travel)
- **Email:** someshchevula25@gmail.com
- **Phone:** +91-9347679388
- **Profiles:**
 - **LinkedIn:** [linkedin.com/in/somesh2504](https://www.linkedin.com/in/somesh2504)
 - **GitHub:** github.com/Somesh2504
 - **LeetCode:** leetcode.com/chevulasomesh
- **Interests & Lifestyle:**
 - **Fitness:** Active in structural strength/weightlifting and endurance running (tracks 21.1km half-marathon distances on Strava).
 - **Sports:** Avid cricket follower, tracking player stats (like Bhuvneshwar Kumar, Travis Head) and team schedules.

2. Education

- **Bachelor of Engineering (B.E.) in Computer Science and Engineering**
 - **Institution:** Matrusri Engineering College, Hyderabad
 - **Timeline:** October 2022 – Expected June 2026
 - **Current CGPA:** 8.38
- **Intermediate (MPC)**
 - **Institution:** TSRJC (BOYS), Sarvail, Telangana
 - **Timeline:** 2020 – 2022
 - **Score:** 978/1000 (97.8%)

3. Technical Skills

- **Programming Languages:** JavaScript (ES6+), TypeScript, Python, Java, SQL, PLpgSQL, HTML5, CSS3.
- **Frontend Technologies:** React.js (v19), Tailwind CSS, Vite, Framer Motion, Socket.IO Client, HTML/XHTML, DOM Manipulation, AJAX.
- **Backend & Databases:** Node.js, Express.js (v5), MongoDB (Mongoose), PostgreSQL, Supabase, Redis.
- **Machine Learning & AI:** PyTorch, HuggingFace, CNNs (ResNet, DenseNet, EfficientNet), XGBoost, Random Forest, SVM, OpenCV, Grad-CAM (Explainable AI).
- **Core CS Fundamentals:** Operating Systems (OS), Computer Networks (CN), Database Management Systems (DBMS), System Design, Data Structures & Algorithms (DSA).
- **Tools & Architecture:** Git, GitHub, Postman, Razorpay API, JWT, Nodemailer, RESTful APIs, WebSockets, WebRTC, MVC Architecture, PM2, Google Colab.

4. Professional Experience

Software Engineer Intern (Frontend & Backend) | Student Tribe (SkyCRM)

Timeline: June 2025 – October 2025

- Architected a 4-tier Customer Relationship Management (CRM) application with role-based access for Admins, Managers, Team Leaders, and Representatives.
- Developed a system for managers to upload bulk leads, dynamically assign them to teams, and track statuses (Interested, Pending, Success, Follow-up).
- Improved dashboard performance using Redis caching to quickly render complex visualizations (bar graphs, pie charts) showing team performance and lead-source ROI.
- Collaborated in a 5-member Agile team to design scalable database schemas and intuitive frontend interfaces.

5. Full-Stack Web Development Projects

Grave / ServeQ (Smart Restaurant Ordering & Queue Management)

- **Tech Stack:** React 19, Vite, Tailwind CSS, Node.js, Express, Supabase (PostgreSQL), Razorpay, JWT.
- **Details:** Built a unified checkout system supporting both UPI (Razorpay with Route transfers/GST calculation) and Cash payments.
- **Key Features:**
 - Implemented atomic token generation using PostgreSQL RPC and row-level locking (FOR UPDATE) to prevent duplicate order numbers under high concurrency.
 - Developed a real-time Kitchen Display System (KDS) and public order tracking page using WebSockets.
 - Created dynamic menu management (soft deletes, availability toggles) and comprehensive admin analytics dashboards for revenue and peak-hour heatmaps.

AlumNet (College Networking Architecture)

- **Tech Stack:** React 19, Node.js, Express, MongoDB, Socket.IO, WebRTC (PeerJS).
- **Details:** A networking platform connecting Students, Alumni, and Colleges with multi-role JWT/OTP authentication.
- **Key Features:**
 - Architected a real-time polymorphic chat and community group messaging system with compound database indexing for rapid queries.
 - Integrated in-browser peer-to-peer audio and video calling using WebRTC and custom socket signaling.
 - Built full CRUD dashboards for colleges and admins, including an automated testimonial moderation system.

6. Machine Learning & Deep Learning Projects

Skin Cancer Diagnostics Architecture (Major College Project)

- **Tech Stack:** Python, PyTorch, CNN Ensembles, XGBoost, Grad-CAM.
- **Details:** A multi-feature fusion framework using the ISIC 2019/HAM10000 datasets to classify skin lesions into 8 diagnostic classes.

- **Key Features:**

- Combined deep semantic features (from DenseNet, ResNet, EfficientNet) with clinical handcrafted features (ABCD rules, GLCM texture metrics).
- Achieved 90.18% overall accuracy and a 94.73% Macro AUC using a soft-voting ensemble (XGBoost, Random Forest, SVM).
- Applied a "safety-first" bias to heavily minimize false negatives for malignant classes, and integrated Grad-CAM heatmaps for Explainable AI (XAI) transparency.

AudioClone (End-to-End Neural Voice Cloning Pipeline)

- **Tech Stack:** Python, PyTorch, Whisper, Demucs, Coqui XTTS v2, Google Colab.
- **Details:** Cloned specific vocal styles (including that of S. P. Balasubrahmanyam) from raw audio files.
- **Key Features:**
 - Built a robust data pipeline to convert MP3s to an LJSpeech-style dataset using Demucs for noise reduction and Silero VAD for audio slicing.
 - Fine-tuned a Coqui XTTS v2 GPT-backbone model, dynamically tuning hyperparameters to fit GPU constraints.
 - Created an interactive ipywidgets UI for real-time text-to-speech generation with adjustable synthesis parameters like speed and temperature.

7. Achievements & Additional Milestones

- **Algorithmic Problem Solving:** Mastered 450+ complex Data Structures and Algorithms (DSA) problems on LeetCode using Java ,participated in various hackthons